about our knowledge of the system rather than about objective frequencies; if somehow we knew exactly what the particle positions were, rather than just their distribution, we could predict experimental outcomes exactly without any need for probabilities at all.

This puts Bohmian mechanics in an interesting position as an alternative formulation of quantum mechanics. GRW theory matches traditional quantum expectations usually, but also makes definite predictions for new phenomena that can be tested. Like GRW, Bohmian mechanics is unambiguously a different physical theory, not simply an "interpretation." It doesn't *have* to obey the Born rule if for some reason our particle positions are not in an equilibrium distribution. But it will obey the rule if they are. And if that's the case, the predictions of Bohmian mechanics are strictly indistinguishable from those of ordinary quantum theory. In particular, we will see more particles hit the screen where the wave function is large, and fewer where it is small.

We still have the question of what happens when we look to see which slit the particle has gone through. Wave functions don't collapse in Bohmian mechanics; as with Everett, they always obey the Schrödinger equation. So how are we supposed to explain the disappearance of the interference pattern in the double-slit experiment?

The answer is "the same way we do in Many-Worlds." While the wave function doesn't collapse, it does evolve. In particular, we should consider the wave function for the detection apparatus as well as for the electrons going through the slits; the Bohmian world is completely quantum, not stooping to an artificial split between classical and quantum realms. As we know from thinking about decoherence, the wave function for the detector will become entangled with that of an electron passing through the slit, and a kind of "branching" will occur. The difference is that the variables describing the apparatus (which aren't there in Many-Worlds) will be at locations corresponding to one of these branches, and not the other. For all intents and purposes, it's just like the wave function has collapsed; or, if you prefer, it's just like decoherence has branched the wave function, but instead of assigning reality to each of the branches, the particles of which we are made are only located on one particular branch.

You won't be surprised to hear that many Everettians are dubious about this kind of story. If the wave function of the universe simply obeys the Schrödinger equation, it will undergo decoherence and branching. And